

BSAVA API Architecture & Interface Specification

Document Metadata	Details
Client	British Small Animal Veterinary Association (BSAVA)
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Version	v1.0.0
Date	August 2026
Status	Technical Baseline
Target Architecture	Composable MACH Ecosystem

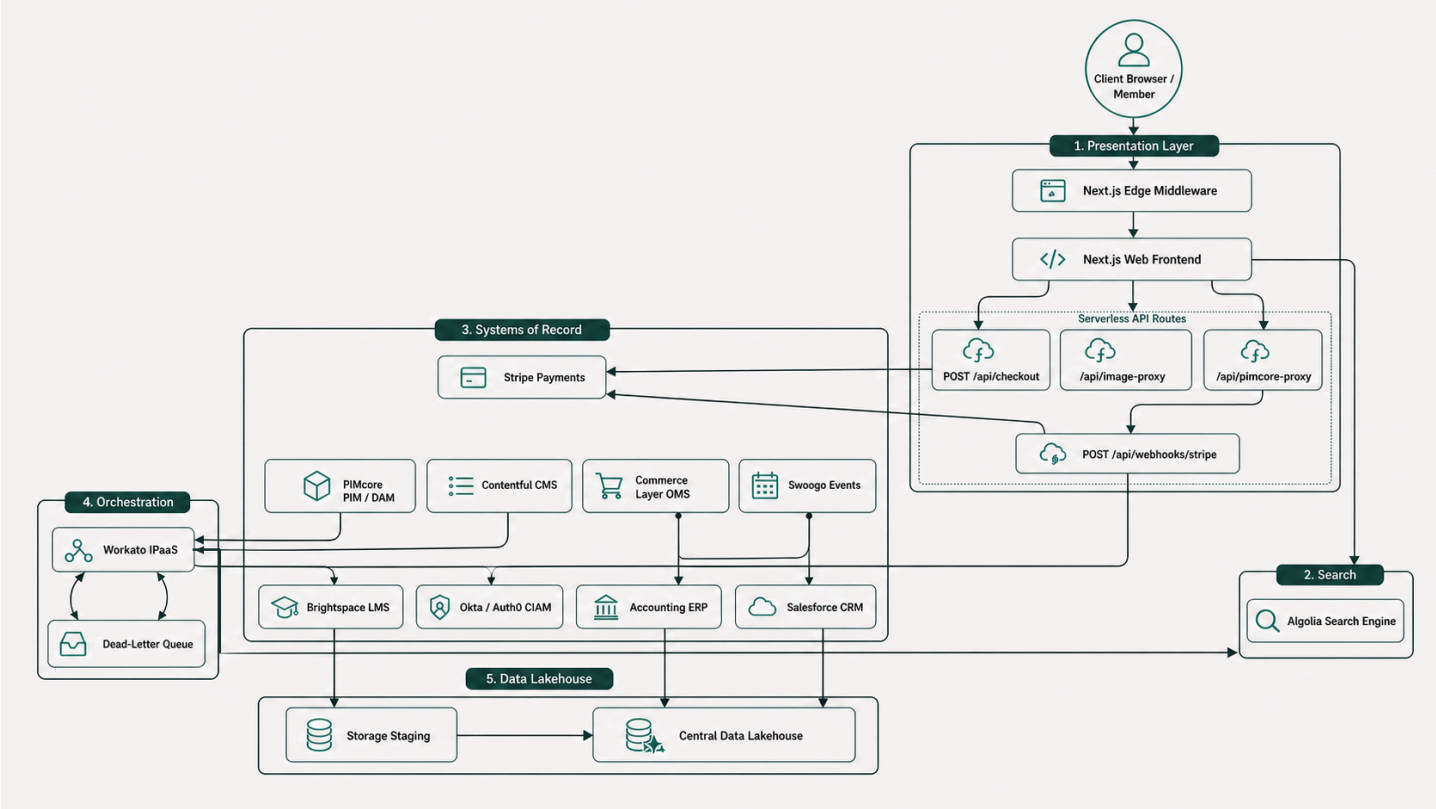
1. Purpose & Scope

This document catalogues every API endpoint, webhook, integration call, and data flow required by the BSAVA MACH platform, organised by **System of Record (SoR)**. It serves as the definitive interface contract for frontend engineers, backend integrators, and Workato recipe developers.

Core Architectural Principles

- Single Source of Truth (SoR):** Every data domain is owned by exactly one operational platform.
- Token-Governed Orchestration:** Workato recipe executions adhere to a **1,000,000 task/year** quota via micro-batching and direct storage staging.
- Edge Security & Privacy:** Next.js Edge Middleware handles JWT authentication and CORS. All PII storage complies with UK GDPR in UK/EU cloud regions.

2. High-Level System Topology



3. API Specification by System of Record

3.1 Salesforce (CRM) — Contact & Member SoR

#	API / Endpoint	Method	Consumer	Purpose
SF-01	REST API — /services/data/vXX.0/subjects/Contact	GET / PATCH	Workato	Query and upsert member contact records
SF-02	REST API — /services/data/vXX.0/subjects/Account	GET / PATCH	Workato	Query and upsert organisation accounts
SF-03	REST API — /services/data/vXX.0/subjects/Committee_Role__c	GET / POST	Workato	Manage committee role assignments
SF-04	REST API — /services/data/vXX.0/subjects/Order	POST	Workato	Create order records from Commerce Layer webhooks
SF-05	REST API — /services/data/vXX.0/query/?q=SELECT...	GET	Workato	Hourly micro-batch query (WHERE LastModifiedDate > {last_run}) for lakehouse ingestion
SF-06	Bulk API 2.0 — /services/data/vXX.0/jobs/ingest	POST	Workato	Mass data load during historic migration (Phase 3)
SF-07	SOAP API	POST	Legacy	Legacy integration support (migration period only)
SF-08	CDC (ChangeEvent) — /data/ContactChangeEvent	Stream	Workato (optional)	Real-time change data capture (high token cost — under review)

Authentication: OAuth 2.0 (Connected App) with refresh token rotation.

Outbound Webhooks from Salesforce:

- CRM-to-CIAM profile sync push to Okta (< 5s latency) on Contact field changes
- Nightly batch reconciliation job for out-of-sync profiles

Lakehouse Ingestion: Hourly scheduled micro-batch → raw_salesforce_contacts (est. 8,760 tasks/year)

3.2 Okta / Auth0 (CIAM) — Identity & Session SoR

#	API / Endpoint	Method	Consumer	Purpose
OK-01	Authentication API — /api/v1/authn	POST	Next.js Frontend	User login, MFA challenge
OK-02	Token Endpoint — /oauth2/default/v1/token	POST	Next.js Frontend	OAuth2 token exchange, refresh (15-min access / 14-day refresh)
OK-03	Authorize Endpoint — /oauth2/default/v1/authorize	GET	Next.js Frontend	SSO redirect flow initiation
OK-04	Management API — /api/v1/users	GET / POST / PUT	Workato	User CRUD, profile sync from Salesforce
OK-05	System Log API — /api/v1/logs	GET	Workato	Security audit log streaming to Data Lakehouse
OK-06	Backchannel Logout — /api/v1/sessions	DELETE	Salesforce / Admin	Global session revocation forcing re-authentication
OK-07	UserInfo Endpoint — /oauth2/default/v1/userinfo	GET	Next.js Frontend	Fetch user profile claims

Authentication: API Token (Management API), OAuth 2.0 PKCE (Frontend flows).

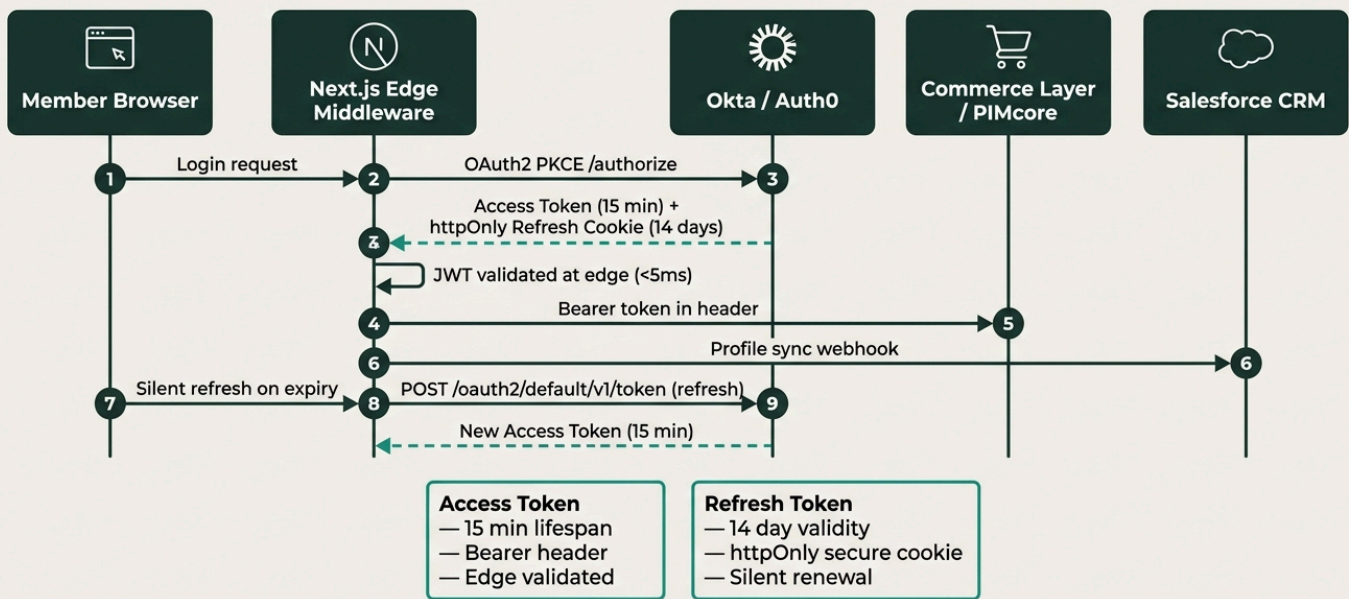
JWT Token Lifecycle:

- Access tokens: 15-minute lifespan, carried as Bearer tokens
- Refresh tokens: httpOnly secure cookie, 14-day validity
- Edge Middleware validates JWT on every request; silent refresh on expiry

Lakehouse Ingestion: System Log API polled daily → security monitoring tables

JWT Token Lifecycle

Okta / Auth0 CIAM — Edge-native session management



3.3 PIMcore (PIM / DAM) — Product Catalog & Entitlements SoR

#	API / Endpoint	Method	Consumer	Purpose
PM-01	GraphQL Data Hub — /pimcore-graphql-webservices/bsava?apikey={key}	POST	Next.js Frontend	Multi-class product listing queries (getBookListing , getEbookListing , getEventListing , getCourseListing , getMembershipTierListing)
PM-02	GraphQL Data Hub — getProduct(id)	POST	Next.js Frontend	Single product detail fetch with inline fragments (Book , Ebook , Event , Course , MembershipTier)
PM-03	GraphQL Data Hub — legacy getProductListing	POST	Next.js Frontend	Fallback query if multi-class schema unavailable
PM-04	REST API — /api/products	POST / PUT	Workato	Create/update event product records (Swoogo → PIMcore sync)
PM-05	REST API — /api/products	POST / PUT	Workato	Create/update course product records (Brightspace → PIMcore sync)
PM-06	REST API — /api/entitlements	GET	Workato / Commerce Layer	Query entitlement definitions and access rules
PM-07	REST API — Asset endpoint	GET	Next.js (via /api/image-proxy)	Serve product images and digital assets

Authentication: Bearer Token (PIMCORE_REST_API_TOKEN) for REST; API Key query parameter for GraphQL Data Hub.

Outbound Webhooks from PIMcore:

Event	Destination	Purpose
product.created	Workato → Algolia	Index new product in search
product.updated	Workato → Algolia	Update search index record
product.deleted	Workato → Algolia	Remove from search index

Lakehouse Ingestion: Nightly cron at 01:00 UTC → snap_pimcore_products (365 tasks/year)

3.4 Contentful (CMS) — Editorial Content SoR

#	API / Endpoint	Method	Consumer	Purpose
CF-01	Content Delivery API (CDA) — getEntries (content_type: article)	GET	Next.js Frontend	Fetch articles by slug, latest articles, all articles for indexing
CF-02	CDA — getEntries (content_type: header)	GET	Next.js Frontend	Fetch site header navigation
CF-03	CDA — getEntries (content_type: footer)	GET	Next.js Frontend	Fetch site footer content
CF-04	CDA — getEntries (content_type: author)	GET	Next.js Frontend	Fetch author profiles and author-article associations
CF-05	Content Preview API — preview.contentful.com	GET	Next.js Frontend	Draft content preview
CF-06	Content Management API (CMA)	PUT / POST	Editorial team	Content authoring (not consumed by frontend)

Authentication: Delivery Token (CDA), Preview Token (Preview API), Management Token (CMA).

Outbound Webhooks from Contentful:

Event	Destination	Purpose
publish	Workato → Algolia	Upsert article into content search index
unpublish	Workato → Algolia	Remove stale article from search index

Lakehouse Ingestion: Nightly cron at 01:00 UTC → snap_contentful_articles (365 tasks/year)

3.5 Commerce Layer (OMS) — Cart, Order & Membership Tier SoR

#	API / Endpoint	Method	Consumer	Purpose
CL-01	REST API — /api/orders	POST	Next.js Frontend	Create order from cart
CL-02	REST API — /api/line_items	POST / DELETE	Next.js Frontend	Add/remove cart items
CL-03	REST API — /api/orders/{id}	PATCH	Next.js Frontend	Update order (apply discounts, customer context)
CL-04	REST API — /api/customers	GET / POST	Workato	Customer record lookup/creation

#	API / Endpoint	Method	Consumer	Purpose
CL-05	REST API — /api/subscriptions	GET	Workato	Query membership subscription tier states
CL-06	REST API — /api/orders?filter[updated_at_gt]={ts}	GET	Workato	Hourly micro-batch order extraction for lakehouse

Authentication: OAuth 2.0 Client Credentials (API credentials).

Outbound Webhooks from Commerce Layer:

Event	Destination	Purpose
order.placed	Workato → Salesforce	Create/update CRM contact and order record
order.placed	Workato → Swoogo	Confirm delegate registration status
order.placed	Workato → Brightspace	Enrol customer in purchased course
order.placed	Workato → ERP	Generate invoice and record payment
subscription.updated	Workato → Salesforce	Sync membership tier changes

Lakehouse Ingestion: Hourly micro-batch → raw_commerce_orders + raw_commercelayer_subscriptions (est. 8,760 tasks/year)

3.6 Stripe — Payment Processing SoR

#	API / Endpoint	Method	Consumer	Purpose
ST-01	Checkout Sessions — stripe.checkout.sessions.create()	POST	Next.js /api/checkout	Create hosted checkout session with line items, bundle discounts, promotion codes
ST-02	PaymentIntents API	POST	Commerce Layer	Initialise payment from OMS order
ST-03	Webhook Signature — stripe.webhooks.constructEvent()	—	Next.js /api/webhooks/stripe	Verify inbound webhook authenticity

Authentication: Secret Key (server-side), Publishable Key (client-side elements).

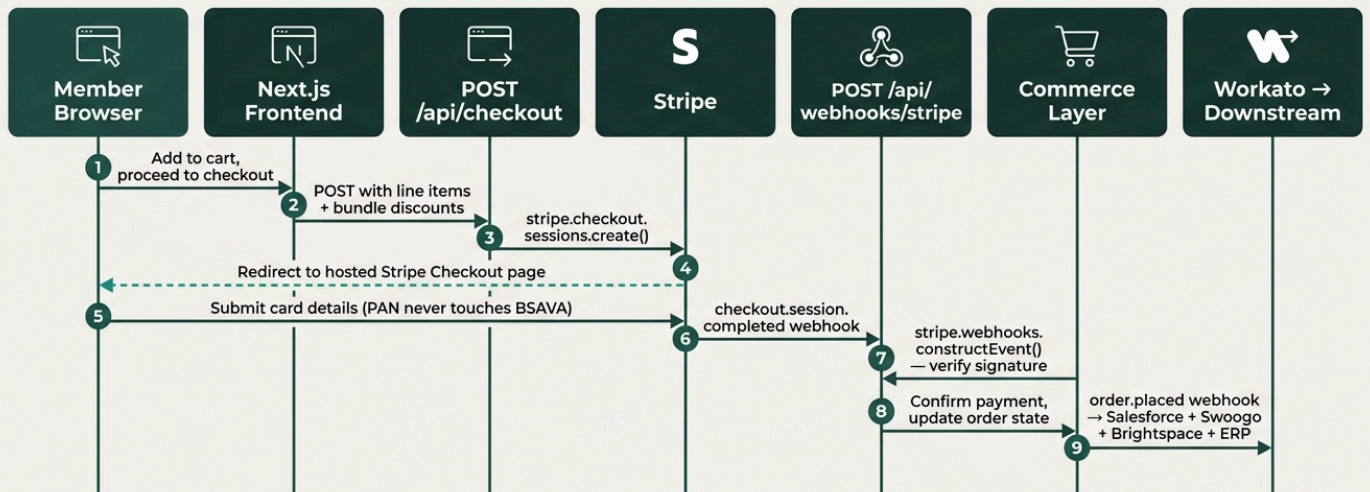
Inbound Webhooks handled by BSAVA:

Event	Handler	Purpose
checkout.session.completed	Next.js /api/webhooks/stripe	Trigger fulfilment logic, log payment
payment_intent.succeeded	Workato	Confirm payment for Commerce Layer order
charge.succeeded	Workato	Financial reconciliation logging

Lakehouse Ingestion: Hourly micro-batch → raw_stripe_payments (combined with Commerce Layer recipe)

Stripe Checkout & Payment Flow

Section 3.6 — PCI SAQ-A scope reduction via hosted checkout



PCI Scope: Card data flows directly browser → Stripe. BSAVA infrastructure qualifies for SAQ A.

3.7 Swoogo — Events & Conferences SoR

#	API / Endpoint	Method	Consumer	Purpose
SW-01	REST API — Event query	GET	Workato	Fetch event details and ticket types
SW-02	REST API — Registration update	PUT	Workato	Mark registration as confirmed after payment
SW-03	REST API — Delegate roster query	GET	Workato	Twice-daily scheduled poll for delegate registration arrays (lakehouse)

Authentication: API Key / OAuth2 (vendor-specific).

Outbound Webhooks from Swoogo:

Event	Destination	Purpose
event.created	Workato → PIMcore	Create event product record (MVE threshold)
event.updated	Workato → PIMcore	Update event product metadata
registration.completed	Workato → Salesforce	Find/create contact, log attendance, assign entitlements

Data Mapping (Swoogo → PIMcore):

Swoogo Field	PIMcore Field
event_name	name
start_date / end_date	eventStartDate / eventEndDate
location	venue
ticket_types[].price	pricing.basePrice
capacity	stockQuantity

Lakehouse Ingestion: Twice-daily poll → raw_swoogo_registrations

3.8 Brightspace / D2L (LMS) — Learning & CPD SoR

#	API / Endpoint	Method	Consumer	Purpose
BS-01	Enrolment API	POST	Workato	Create course enrolment after product purchase
BS-02	Data Hub — Bulk CSV export	GET	Workato	Daily pull of course completions and CPD credit files
BS-03	Course API — Enrolled courses	GET	Next.js Frontend	Fetch member's enrolled courses and progress %
BS-04	SSO Launcher	GET	Next.js Frontend	Generate signed SSO URL for course player redirect
BS-05	Course Metadata API	GET	Workato	Fetch course title, description, syllabus, CPD credits for PIMcore sync

Authentication: OAuth 2.0 / LTI (Learning Tools Interoperability) for SSO.

Outbound Events from Brightspace (ENS):

Event	Destination	Purpose
course.created	Workato → Algolia	Index new course in search
course.updated	Workato → PIMcore	Update course product metadata
course.updated	Workato → Algolia	Update course search record
course.completed	Workato → Salesforce	Record CPD completion, trigger renewal flows

Lakehouse Ingestion: Daily Data Hub CSV → S3/ADLS staging → raw_brightspace_completions (365 tasks/year)

3.9 Accounting ERP — Financial Ledger SoR

#	API / Endpoint	Method	Consumer	Purpose
ERP-01	REST API — Invoice creation	POST	Workato	Create invoice and record payment from Commerce Layer order
ERP-02	REST API — Journal entry	POST	Workato	Post general ledger journal entries
ERP-03	REST API / Scheduled export — GL query	GET	Workato	Extract GL entries and monthly trial balances for lakehouse

Authentication: API Key / OAuth2 (vendor-specific).

Lakehouse Ingestion: Nightly scheduled export → S3/ADLS staging → raw_erp_gl_entries

3.10 Algolia — Search Index (Operational Cache)

#	API / Endpoint	Method	Consumer	Purpose
AL-01	Search Client SDK — products index	GET	Next.js Frontend (client-side)	Instant product and event search
AL-02	Search Client SDK — content index	GET	Next.js Frontend (client-side)	Instant article and editorial search
AL-03	Admin API — Upsert record	POST / PUT	Workato	Add/update product, event, course, or article in search index
AL-04	Admin API — Delete record	DELETE	Workato	Remove unpublished or deleted records from index

Authentication: Search API Key (client-side, read-only), Admin API Key (server-side, write).

Index Sources:

Index	Source System	Sync Trigger
products	PIMcore	Webhook on product create/update/delete
content	Contentful	Webhook on publish/unpublish
content (courses)	Brightspace	Webhook on course create/update

3.11 Data Lakehouse — Enterprise Analytics

#	API / Endpoint	Method	Consumer	Purpose
LH-01	Bulk stage loading (Snowpipe / ADLS connector)	PUT	Workato	Land JSON/CSV/Parquet batches from operational systems
LH-02	Direct storage copy (S3 / ADLS / OneLake)	File transfer	Workato	Bulk file landing for datasets > 10,000 records
LH-03	SQL endpoint (JDBC / ODBC)	SELECT	Power BI / Tableau	BI queries against Silver/Gold zone tables

Target Tables:

Table	Source	Ingestion Pattern
raw_salesforce_contacts	Salesforce	Hourly micro-batch
raw_commercelayer_subscriptions	Commerce Layer	Hourly micro-batch
raw_commerce_orders	Commerce Layer	Hourly micro-batch
raw_stripe_payments	Stripe	Hourly micro-batch
raw_swoogo_registrations	Swoogo	Twice-daily poll
raw_brightspace_completions	Brightspace	Daily CSV via Data Hub
raw_erp_gl_entries	ERP	Nightly export
snap_pimcore_products	PIMcore	Nightly snapshot (01:00 UTC)
snap_contentful_articles	Contentful	Nightly snapshot (01:00 UTC)

4. Next.js Frontend API Routes (Implemented)

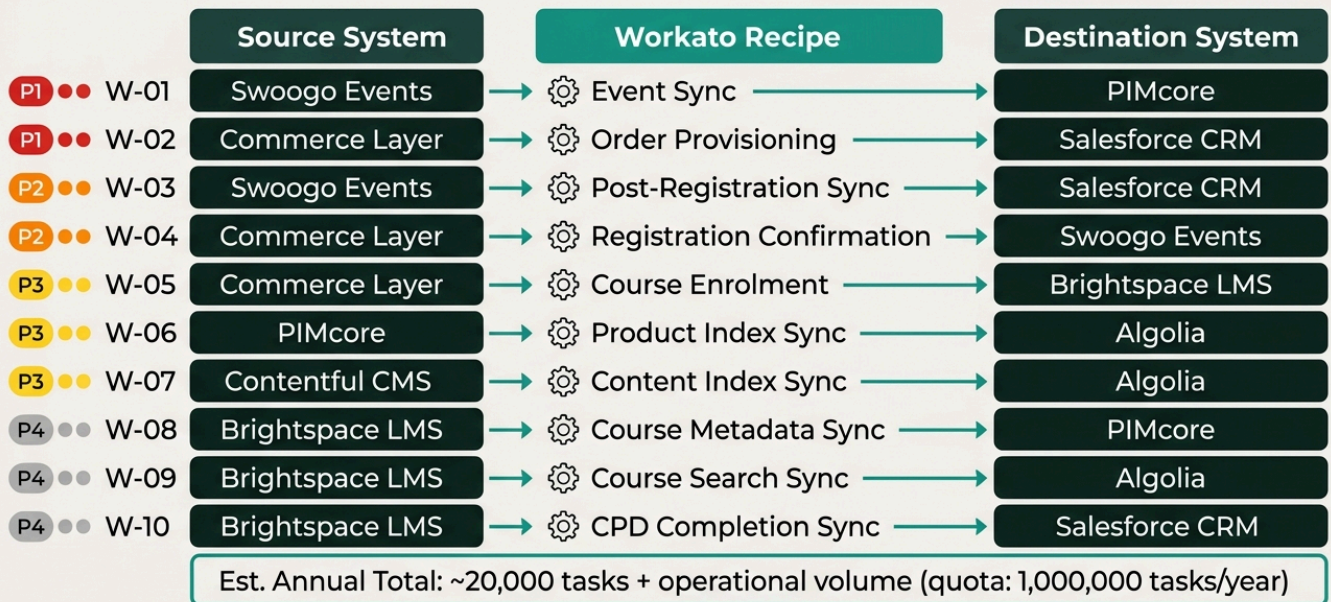
#	Route	Method	Purpose	Upstream System
FE-01	/api/checkout	POST	Create Stripe checkout session with line items and bundle discounts	Stripe
FE-02	/api/webhooks/stripe	POST	Handle checkout.session.completed webhook, verify signature	Stripe
FE-03	/api/pimcore-proxy	POST	Proxy GraphQL queries to PIMcore Data Hub	PIMcore
FE-04	/api/image-proxy	GET	Proxy PIMcore asset images to frontend	PIMcore

5. Workato Orchestration Recipes Summary

#	Recipe	Trigger	Source → Destination	Priority
W-01	Event Sync	event.created / event.updated webhook	Swoogo → PIMcore	P1
W-02	Order Provisioning	order.placed webhook	Commerce Layer → Salesforce	P1
W-03	Post-Registration Sync	registration.completed webhook	Swoogo → Salesforce	P2
W-04	Registration Confirmation	order.placed webhook	Commerce Layer → Swoogo	P2
W-05	Course Enrolment	order.placed webhook (if courseId metadata)	Commerce Layer → Brightspace	P3
W-06	Product Index Sync	Product webhook	PIMcore → Algolia	P3
W-07	Content Index Sync	publish / unpublish webhook	Contentful → Algolia	P3
W-08	Course Metadata Sync	course.updated / nightly job	Brightspace → PIMcore	P4
W-09	Course Search Sync	course.created / course.updated webhook	Brightspace → Algolia	P4
W-10	CPD Completion Sync	course.completed event	Brightspace → Salesforce	P4

Workato Orchestration Recipe Map

W-01 to W-10 — Trigger, source and destination by priority



Estimated Annual Token Budget

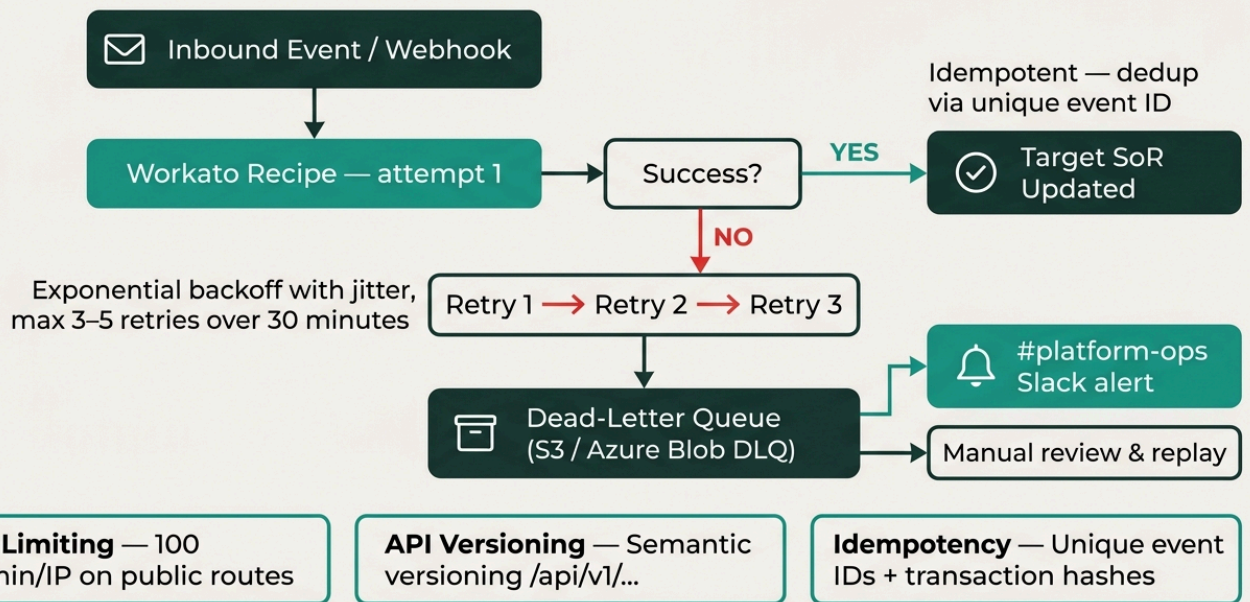
Recipe Group	Pattern	Est. Tasks/Year
Member Profile & Subscription Ingestion	Hourly micro-batch	~8,760
Financial Transaction & Order Ingestion	Hourly micro-batch	~8,760
Event Delegate Analytics	Twice-daily poll	~730
Learning & CPD Credit Ingestion	Daily bulk pull	~365
Product & Content Catalog Snapshots	Nightly cron	~730
Operational webhook recipes (W-01 to W-10)	Event-driven	Variable
Estimated Total		~20,000 + operational volume

6. Error Handling & Resilience

Pattern	Detail
Retry Policy	Exponential backoff with jitter, max 3–5 retries over 30 minutes
Dead-Letter Queue	Failed payloads written to S3/Azure Blob DLQ; #platform-ops Slack alert
Idempotency	All consumers enforce dedup via unique event IDs / transaction hashes
Rate Limiting	Public API routes: 100 req/min/IP. Downstream systems enforce API token quotas
API Versioning	All custom service interfaces use semantic versioning (/api/v1/...)

Error Handling & Resilience

Retry, Dead-Letter Queue and idempotency patterns



7. Authentication Summary

Authentication Summary

Auth methods and token types by system



Salesforce

Workato

Auth Method	OAuth 2.0 Connected App
Token / Key Type	Access Token + Refresh Token



Okta / Auth0

Next.js

Auth Method	OAuth 2.0 PKCE (frontend) / API Token (mgmt)
Token	JWT Access + httpOnly Refresh Cookie



PIMcore

Frontend

Auth Method	Bearer Token (REST) / API Key (GraphQL)
Token	PIMCORE_REST_API_TOKEN



Contentful

Frontend

Auth Method	API Tokens
Token / Key Type	Delivery / Preview / Management tokens



Commerce Layer

Next.js

Auth Method	OAuth 2.0 Client Credentials
Token	Client ID + Secret



Stripe

Frontend

Auth Method	API Keys
Token / Key Type	Secret Key (server) / Publishable Key (client)



Swoogo

Frontend

Auth Method	API Key / OAuth2 (vendor)
Token / Key Type	Vendor-specific



Brightspace

Frontend

Auth Method	OAuth 2.0 / LTI
Token / Key Type	SSO launcher tokens



Algolia

Frontend

Auth Method	API Keys
Token / Key Type	Search Key (read-only) / Admin Key (write)

System	Auth Method	Token/Key Type
Salesforce	OAuth 2.0 Connected App	Access Token + Refresh Token
Okta / Auth0	OAuth 2.0 PKCE (frontend), API Token (management)	JWT Access + httpOnly Refresh Cookie
PIMcore	Bearer Token (REST), API Key (GraphQL)	PIMCORE_REST_API_TOKEN
Contentful	API Tokens	Delivery, Preview, Management tokens

System	Auth Method	Token/Key Type
Commerce Layer	OAuth 2.0 Client Credentials	Client ID + Secret
Stripe	API Keys	Secret Key (server), Publishable Key (client)
Swoogo	API Key / OAuth2	Vendor-specific
Brightspace	OAuth 2.0 / LTI	SSO launcher tokens
Algolia	API Keys	Search Key (read), Admin Key (write)

End of API Architecture Document.